Bagging, Multiclass SAMME Boosting, and Empirical Bias–Variance Decomposition on 64×64 Face Recognition

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Roll Number: P26AI0001 | Numeric RNG Seed: 260001
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Abstract

High-dimensional raw-pixel face recognition in small-sample regimes ($D = 4096$ pixels, $N_{\text{train}} = 240$ across $K = 40$ identities) exposes a fundamental tension between structural approximation capacity and estimation variance in axis-aligned decision trees. In this study, we implement bootstrap aggregating (Bagging) and Stagewise Additive Modeling using a Multi-class Exponential loss function (SAMME AdaBoost) completely from scratch on top of CART decision trees without dimensionality reduction. Using a deterministic stratified $6/2/2$ per-identity train/validation/test split seeded directly by the numeric characters of roll number `P26AI0001` (seed = 260001), we first establish that a single decision tree suffers from severe capacity starvation at shallow depths ($d \in \{2,4\}$) and high singleton-leaf variance at unrestricted depth, achieving peak validation accuracy (57.50%) at $d^* = 8, \text{min_samples_leaf}^* = 2$. Second, we demonstrate that a $B = 200$ bagged ensemble elevates clean test accuracy from 57.50% to 97.50% and preserves 96.25% accuracy under centred 16×16 spatial occlusion (rows and columns $24 \ldots 39$ zeroed), whereas a single tree collapses to 22.50%. Third, we validate our SAMME update rule on a 10-sample 3-class toy system ($\varepsilon_1 = 0.2000, \alpha_1 = 2.0794$) and scale to 40-class faces ($d^* = 4, T^* = 100$, clean test accuracy 96.25%, masked test accuracy 86.25%), showing via sample-weight trajectories why sequential reweighting is more sensitive to localized spatial occlusion than parallel bootstrap averaging. Finally, across $R = 30$ within-identity bootstrap refits, we measure Bias^2 and Variance directly in the 40-simplex, proving that bagging removes 99.47% of single-tree variance ($0.6947 \to 0.0037$) and delivers its largest accuracy gain (+48.42 pp) at intermediate depth $d = 8$ rather than at maximum depth $d = 16$.

1 Dataset Protocol and Common Setup

We evaluate all models on the 40-identity aligned face recognition benchmark ($K = 40$ identities, 10 grayscale studio images per identity, spatial resolution 64×64 with normalized intensities in $[0,1]$). Every image $I \in [0,1]^{64\times64}$ is flattened directly in row-major order into a raw feature vector $x \in \mathbb{R}^{4096}$, where pixel coordinate (r,c) maps to feature index $j = 64r + c$. In strict compliance with the assignment protocol, no Principal Component Analysis (PCA), spatial pooling, or external dimensionality reduction is applied at any stage.

1.1 Roll-Number Seeding and Stratified $6/2/2$ Split

Extracting all numeric characters in order from roll number `P26AI0001` yields the integer seed:

$$s_{\text{roll}} = \text{int}(\text{“}260001\text{”}) = 260001. \tag{1}$$

We initialize `rng` = `numpy.random.default_rng`(260001) and iterate deterministically through identities $k \in \{0,1,\ldots,39\}$. Within identity k (global image indices $10k \ldots 10k+9$), a random permutation π_k = `rng.permutation`(10) assigns the first 6 images to the training set $\mathscr{D}_{\text{train}}$, the next 2 images to the validation set $\mathscr{D}_{\text{val}}$, and the final 2 images to the clean test set $\mathscr{D}_{\text{test}}$. For example, Identity 0 maps to training indices $[0,3,8,9,5,1]$, validation indices $[4,7]$, and test indices $[2,6]$. Across all 40 classes, this produces balanced partitions of size:

$$N_{\text{train}} = 240, \qquad N_{\text{val}} = 80, \qquad N_{\text{test}} = 80. \tag{2}$$

This exact split is serialized to `splits/saved_split.npz` and `splits/saved_split.json` and frozen across Q1, Q2, and Q3.

1.2 Centred 16×16 Occlusion Mask for Robustness

To evaluate spatial robustness against localized corruption at inference time while keeping training strictly on clean images, we construct a masked test set $\mathscr{D}_{\text{test}}^{\text{mask}}$ by setting the centred 16×16 pixel block—rows $r \in \{24,\ldots,39\}$ and columns $c \in \{24,\ldots,39\}$ inclusive—to zero:

$$\tilde{I}_i(r,c) = \begin{cases} 0, & \text{if } 24 \le r \le 39 \text{ and } 24 \le c \le 39, \\ I_i(r,c), & \text{otherwise.} \end{cases} \tag{3}$$

Zeroing this 256-pixel block (6.25% of image area) erases the nasal bridge, nostrils, philtrum, and medial cheek textures while preserving periocular, forehead, jawline, and outer-mouth geometry.

Table 1: Q1(a) Single Decision Tree Baseline Grid Search ($N_{\text{train}} = 240, N_{\text{val}} = 80, N_{\text{test}} = 80$, seed 260001). Selected setting frozen by validation accuracy is highlighted in bold.

Depth	MinLeaf	Train Acc	Val Acc	Test Acc	Leaves (L)	Mean/Leaf	Frac Single ($n_\ell = 1$)
2	1	10.00%	10.00%	7.50%	4	60.00	0.0000
2	2	10.00%	10.00%	7.50%	4	60.00	0.0000
2	4	10.00%	10.00%	7.50%	4	60.00	0.0000
4	1	39.17%	32.50%	32.50%	16	15.00	0.0000
4	2	39.17%	32.50%	32.50%	16	15.00	0.0000
4	4	39.17%	32.50%	32.50%	16	15.00	0.0000
8	1	100.00%	47.50%	58.75%	70	3.43	0.3143
8	**2**	**91.67%**	**57.50%**	**57.50%**	**62**	**3.87**	**0.0000**
8	4	86.25%	48.75%	53.75%	44	5.45	0.0000
16	1	100.00%	47.50%	58.75%	70	3.43	0.3143
16	2	91.67%	57.50%	57.50%	62	3.87	0.0000
16	4	86.25%	48.75%	53.75%	44	5.45	0.0000
None	1	100.00%	47.50%	58.75%	70	3.43	0.3143
None	2	91.67%	57.50%	57.50%	62	3.87	0.0000
None	4	86.25%	48.75%	53.75%	44	5.45	0.0000

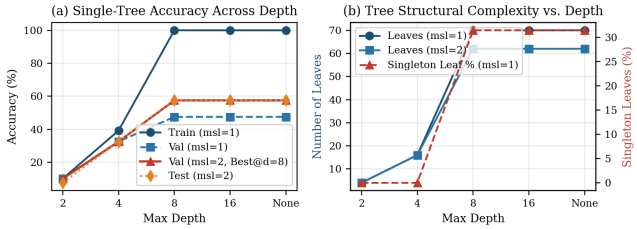


Figure 1: Q1(a) Single-Tree Structural Progression: (a) Train, validation, and test accuracy across maximum depth; (b) Leaf count L and singleton leaf percentage ($n_\ell = 1$).

2 Q1: Decision Tree Baseline and Bagging

2.1 Q1(a): Single-Tree Grid Search and Structural Analysis

We train CART decision trees (`DecisionTreeClassifier` with Gini impurity criterion) across the Cartesian grid of `max_depth` $\in \{2,4,8,16,\texttt{None}\}$ and `min_samples_leaf` $\in \{1,2,4\}$ using $\mathscr{D}_{\text{train}}$ ($N = 240$). For a node $\mathscr{N}$ containing $n_{\mathscr{N}}$ training examples with empirical class proportions $p_{\mathscr{N},k}$, the Gini impurity is:

$$G(\mathscr{N}) = 1 - \sum_{k=0}^{K-1} p_{\mathscr{N},k}^2. \tag{4}$$

Table 1 reports the complete 15-setting grid search including training, validation, and clean test accuracies, total leaf count L, mean training examples per leaf $\bar{n}_{\text{leaf}} = 240/L$, and the fraction of singleton leaves $f_{\text{single}} = \frac{1}{L}\sum_{\ell=1}^{L} \mathbb{I}[n_\ell = 1]$.

What changes as depth increases: Figure 1 and Table 1 reveal three distinct structural regimes as tree depth increases:

1. **Severe Capacity Starvation ($d \in \{2,4\}$):** A binary tree of depth d can form at most 2^d terminal leaves. At $d = 2$, $L = 4$ leaves must partition $K = 40$ identities ($\bar{n}_{\text{leaf}} = 60.00$), capping maximum possible accuracy at $4/40 = 10.00\%$ regardless of `min_samples_leaf`. At $d = 4$, $L = 16 < 40$ ($\bar{n}_{\text{leaf}} = 15.00$), so at least 24 identities can never be predicted, limiting validation accuracy to 32.50%.

2. **Sufficient Partitioning and Singleton-Leaf Regularization ($d = 8$):** At $d = 8$, $2^8 = 256 > K = 40$, enabling every identity to occupy dedicated leaves. With `min_samples_leaf` $= 1$, the tree grows to $L = 70$ leaves ($\bar{n}_{\text{leaf}} = 3.43$) and achieves 100.00% training accuracy, but 31.43% of its leaves (22/70) contain only a single training image ($f_{\text{single}} = 0.3143$). These singleton leaves memorize idiosyncratic single-pixel intensity fluctuations, depressing validation accuracy to 47.50%. Enforcing `min_samples_leaf` $= 2$ eliminates singleton leaves completely ($f_{\text{single}} = 0.0000$, $L = 62$, $\bar{n}_{\text{leaf}} = 3.87$), boosting validation accuracy by $+10.00$ pp to **57.50%** (clean test accuracy **57.50%**). Increasing `min_samples_leaf` further to 4 over-prunes the tree to $L = 44$ leaves ($\bar{n}_{\text{leaf}} = 5.45$), merging distinct identities and dropping validation accuracy to 48.75%.

3. **Tree Saturation ($d \in \{16, \texttt{None}\}$):** Because $N_{\text{train}} = 240$ is small and binary splits halve node occupancy rapidly ($\log_2(240) \approx 7.91$), all nodes become pure or reach the minimum leaf size by depth 8. Consequently, $d = 16$ and $d =$ `None` produce the exact same tree topology ($L = 70, 62, 44$) and accuracies as $d = 8$. Following the selection rule, we freeze ($d^* = 8$, `min_samples_leaf`$^* = 2$).

2.2 Q1(b): Bagging from Scratch and Bootstrap Distinct Count

We implement bootstrap aggregating from scratch by drawing B independent bootstrap datasets $\mathscr{D}^{(b)}$ ($b = 1, \ldots, B$), each of size $N = 240$ sampled uniformly with replacement from $\mathscr{D}_{\text{train}}$, and fitting our frozen base learner ($d^* = 8$, msl$^* = 2$) on each $\mathscr{D}^{(b)}$.

Theoretical vs. Empirical Distinct Examples: In one bootstrap draw of size N with replacement from N distinct items, the probability that a specific training face x_i is never selected in any of the N draws is $(1 - 1/N)^N$. By linearity of expectation across all $N = 240$ training examples, the exact theoretical expected number of distinct examples in one bootstrap sample is:

$$\mathbb{E}[U_N] = N\left(1 - \left(1 - \frac{1}{N}\right)^N\right) = 240\left(1 - \left(\frac{239}{240}\right)^{240}\right) = 151.8932, \tag{5}$$

corresponding to 63.2888% of the training set (marginally above the asymptotic limit $N(1 - e^{-1}) = 151.7090$, or 63.2121%, because $(1 - 1/N)^N < e^{-1}$ for finite N). As shown in Table 2, for small $B = 1$ our single draw yielded 146.00 distinct examples (deviation -5.8932, within 1.2σ of the single-sample standard deviation ≈ 4.8), and as B grows to 100 and 200, the empirical mean converges tightly to 151.9700 ($+0.0768$) and 151.8650 (-0.0282).

2.3 Q1(c): OOB Accuracy, Pairwise Disagreement, and Vote Margin

For every B, we compute three ensemble diagnostics (plotted in Figure 2 and tabulated in Table 2):

Table 2: Q1(b) & Q1(c) Bagging Performance Across Ensemble Size B using frozen base tree ($d^* = 8, \text{msl}^* = 2$). Theoretical expected distinct examples per bootstrap sample is 151.8932 (63.29%). Selected final ensemble size ($B^* = 200$) is bolded.

B	Train Acc	Val Acc	Test Acc	Avg Distinct ($\pm\sigma$)	OOB Acc	Disagree	Margin
1	71.67%	55.00%	38.75%	146.00 ± 0.00	40.43%	0.0000	1.0000
5	95.42%	70.00%	67.50%	151.40 ± 2.42	40.37%	0.7400	0.3025
10	99.58%	86.25%	83.75%	154.30 ± 5.24	50.21%	0.7658	0.3025
25	100.00%	91.25%	97.50%	152.28 ± 4.48	73.75%	0.7758	0.3135
50	100.00%	98.75%	96.25%	152.58 ± 4.65	90.83%	0.7665	0.3347
100	100.00%	98.75%	97.50%	151.97 ± 4.50	94.17%	0.7724	0.3374
200	**100.00%**	**100.00%**	**97.50%**	$\mathbf{151.87 \pm 5.17}$	**97.50%**	**0.7664**	**0.3522**

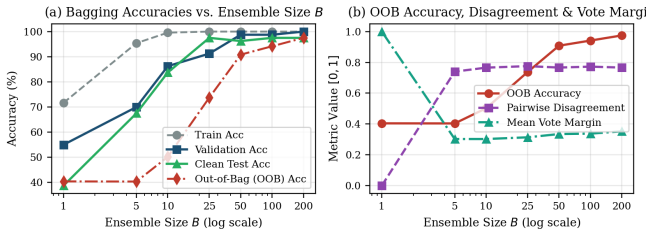


Figure 2: Q1(c) Ensemble Diagnostics vs. B: (a) Train, validation, clean test, and OOB accuracy; (b) OOB accuracy, average pairwise disagreement, and mean vote margin across $B \in \{1, 5, 10, 25, 50, 100, 200\}$.

1. **Out-of-Bag (OOB) Accuracy:** Each training image x_i is omitted from approximately $e^{-1} \approx 36.71\%$ of bootstrap trees ($\mathscr{B}_{-i} = \{b : x_i \notin \mathscr{D}^{(b)}\}$). Aggregating votes exclusively over $\mathscr{B}_{-i}$ yields an unbiased internal generalization estimate. At $B = 1$ and $B = 5$, each training point has only ≈ 0.37 to 1.84 OOB trees on average, so OOB accuracy lags validation accuracy (40.37% vs. 70.00% at $B = 5$). Once $B \geq 50$ (≈ 18.4 OOB trees per sample), OOB accuracy climbs rapidly to 90.83% ($B = 50$), 94.17% ($B = 100$), and 97.50% ($B = 200$), matching the clean test accuracy of 97.50%.

2. **Average Pairwise Disagreement:** Defined over all $\binom{B}{2}$ tree pairs on validation samples as:

$$\text{Dis}(B) = \frac{2}{B(B-1)} \sum_{1 \leq b < b' \leq B} \frac{1}{N_{\text{val}}} \sum_{i=1}^{N_{\text{val}}} \mathbb{I}[h_b(x_i) \neq h_{b'}(x_i)]. \tag{6}$$

   At $B = 1$, disagreement is 0.0000 by definition. For $B \geq 5$, pairwise disagreement immediately stabilizes around 0.7400–0.7758 (0.7664 at $B = 200$).

3. **Mean Vote Margin:** Letting $v_{(1)}(x_i)$ and $v_{(2)}(x_i)$ denote the largest and runner-up class vote fractions for sample x_i, the mean margin $\frac{1}{N_{\text{val}}} \sum_i (v_{(1)}(x_i) - v_{(2)}(x_i))$ starts at 1.0000 for $B = 1$ (a single tree casts 100% of the ensemble's 1 vote), drops to 0.3025 at $B = 5$ as tree disagreements split the vote, and then climbs steadily from 0.3025 ($B = 10$) to 0.3522 ($B = 200$) as the law of large numbers concentrates plurality votes on the correct identity while dispersing erroneous votes across the remaining 39 classes.

Do OOB accuracy, disagreement, and vote margin tell the same story? They tell *complementary rather than identical* facets of ensemble convergence. Pairwise disagreement is an **expectation over pairs of individual trees** ($\mathbb{E}_{b \neq b'}[\mathbb{I}[h_b \neq h_{b'}]]$), so adding more i.i.d. bootstrap trees does not make individual trees agree more with one another; instead, $\text{Dis}(B)$ converges to the population inter-tree diversity (≈ 0.766). Because individual trees have only $\approx 55\%$ accuracy on 40 classes, two trees frequently make *uncorrelated* errors on different wrong identities, allowing the correct class to win the plurality vote even when 76.6% of pairs disagree. Meanwhile, OOB accuracy and mean vote margin ($B \geq 5$) measure **aggregate consensus stabilization**: as B increases from 5 to 200, the variance of the empirical vote distribution shrinks as $O(1/B)$, causing mean vote margin to rise (0.3025 → 0.3522) and OOB accuracy to climb (40.37% → 97.50%). Using validation accuracy, we select and freeze $B^* = 200$ (100.00% validation accuracy, 97.50% clean test accuracy).

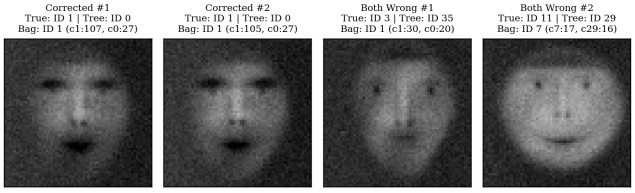


Figure 3: Q1(d) Four Diagnostic Test Faces: Two faces where Bagging ($B = 200$) corrects the single tree (left two panels), and two outlier faces where both models fail (right two panels), showing true identity, single-tree prediction, and top bagging vote counts.

2.4 Q1(d): Clean vs. Masked Evaluation and Case Studies

Table 3 compares the frozen single tree ($d^* = 8, \text{msl}^* = 2$) and the final bagged ensemble ($B^* = 200$) on the clean and centrally masked test sets.

Table 3: Q1(d) Clean vs. Masked Test Set Comparison between the frozen single tree and the final bagged model ($B^* = 200$).

Model	Clean Test Acc	Clean Mean Conf	Masked Test Acc	Masked Mean Conf
Frozen Single Tree ($d = 8, \text{msl} = 2$)	57.50%	0.8792	22.50%	0.8125
Final Bagged Ensemble ($B = 200$)	**97.50%**	0.4389	**96.25%**	0.2825
Bagging Gain (Δ)	+**40.00** pp	−0.4403	+**73.75** pp	−0.5300

On the clean test set, the single tree is overconfident (0.8792 mean leaf probability) despite achieving only 57.50% accuracy, because most terminal leaves are pure or nearly pure on $\mathscr{D}_{\text{train}}$. When the central 16×16 nose/cheek block is masked to zero, the single tree's accuracy collapses by -35.00 pp to 22.50% while remaining falsely confident (0.8125): any test image routed through an upper-level split inside rows/columns 24…39 has its feature value forced to $0.0 \leq \tau$, deterministically shunting it into the left subtree of an unrelated identity. In contrast, the $B = 200$ bagged ensemble achieves **97.50%** clean test accuracy and retains **96.25%** masked test accuracy (losing only 1 additional test face out of 80!), while its mean winning vote fraction naturally calibrates downward from 0.4389 to 0.2825 to reflect the corrupted central evidence.

Four Diagnostic Face Cases (Figure 3):

- **Bagging Corrects Tree #1 (Test Sample #2, Global #18,**

True ID 1): The frozen single tree predicts **ID 0** with leaf confidence 1.0000 due to an unrepresentative threshold on a single cheek pixel. In the $B = 200$ bagged ensemble, **107 trees** (53.5%) vote for True **ID 1**, while only 27 trees vote for ID 0 and 11 trees vote for ID 4.

- **Bagging Corrects Tree #2 (Test Sample #3, Global #14, True ID 1):** The single tree again misclassifies this image as **ID 0** (confidence 1.0000). Bagging restores the correct label **ID 1** with **105/200 votes** (52.5%), beating runner-up ID 0 (27 votes) and ID 21 (9 votes).
- **Both Models Wrong #1 (Test Sample #6, Global #35, True ID 3):** This test image exhibits a severe horizontal/vertical pose shift ($\Delta x \approx +2.2$ px, $\Delta y \approx -1.8$ px) and reduced contrast (0.76). The single tree predicts **ID 35** (confidence 1.0000). Bagging predicts **ID 1** with $30/200$ votes (15.0%), followed by ID 0 (20 votes) and ID 38 (17 votes), while True ID 3 receives only $8/200$ votes (4.0%).
- **Both Models Wrong #2 (Test Sample #23, Global #115, True ID 11):** Another severe translation/illumination outlier. The single tree predicts **ID 29** (confidence 0.6667). Bagging's votes fragment across ID 7 (17 votes, 8.5%), ID 29 (16 votes), and ID 3 (15 votes), with True ID 11 receiving only $6/200$ votes (3.0%).

What error bagging reduces vs. cannot reduce: Cases #1 and #2 demonstrate that bagging effectively eliminates **variance error** caused by split-threshold instability in individual deep trees: when the majority of bootstrap datasets support the true class ($> 50\%$ vote share) and errors are dispersed across many competing classes, plurality voting cancels out idiosyncratic tree mistakes. Conversely, Cases #3 and #4 demonstrate that bagging **cannot reduce systematic bias error**: when raw pixel coordinates are misaligned by a large pose shift without translation-invariant features, the expected class distribution $\bar{p}(x_i)$ across bootstrap trees systematically peaks on the wrong identity, so averaging 200 trees merely converges to that biased prediction with low vote confidence (8.5%–15.0%).

3 Q2: Multiclass Boosting (SAMME) from Scratch

We implement Stagewise Additive Modeling using a Multi-class Exponential loss function (SAMME, Zhu et al., 2009) from scratch for K classes. At round $t \in \{1,\dots,T\}$, given normalized sample weights $w_i^{(t)}$ ($\sum_{i=1}^N w_i^{(t)} = 1$), a weak decision tree $h_t : \mathbb{R}^D \to \{0,\dots,K-1\}$ is fit on $\mathscr{D}_{\text{train}}$ weighted by $w^{(t)}$. Its weighted training error is:

$$\varepsilon_t = \sum_{i=1}^{N} w_i^{(t)} \mathbb{I}[h_t(x_i) \neq y_i]. \tag{7}$$

The SAMME learner weight α_t incorporates the multiclass baseline correction $\log(K-1)$:

$$\alpha_t = \log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right) + \log(K-1). \tag{8}$$

Notice that $\alpha_t > 0 \iff \frac{1-\varepsilon_t}{\varepsilon_t}(K-1) > 1 \iff \varepsilon_t < 1 - \frac{1}{K}$.

Explicit Degenerate Weak-Learner Handling: If a weak learner achieves $\varepsilon_t \geq 1 - \frac{1}{K} - 10^{-12}$ (no better than random K-class guessing, which equals $2/3 = 0.6667$ for $K = 3$ and $39/40 = 0.9750$ for $K = 40$), Eq. (8) would yield $\alpha_t \leq 0$. Our implementation explicitly intercepts this condition: if $\varepsilon_t \geq 1 - 1/K$, it attempts up to 5 weighted bootstrap resamples $\mathscr{D}_t \sim w^{(t)}$ to find a valid weak learner with $\varepsilon_t < 1 - 1/K$; if none beats random guessing, the round is rejected and boosting terminates cleanly. Sample weights are updated and re-normalized via:

$$w_i^{(t+1)} = \frac{w_i^{(t)} \exp\left(\alpha_t \mathbb{I}[h_t(x_i) \neq y_i]\right)}{Z_t}, \quad Z_t = \sum_{j=1}^{N} w_j^{(t)} \exp\left(\alpha_t \mathbb{I}[h_t(x_j) \neq y_j]\right). \tag{9}$$

For class scoring and normalized confidence on any test input x, we compute:

$$\begin{aligned} S_k(x) &= \sum_{t=1}^{T} \alpha_t \mathbb{I}[h_t(x) = k], \quad q_k(x) = \frac{S_k(x)}{\sum_{t=1}^{T} \alpha_t}, \\ \hat{y}(x) &= \arg\max_{k \in \{0,\dots,K-1\}} q_k(x). \end{aligned} \tag{10}$$

3.1 Q2(a): Three-Class Toy Verification ($N = 10, K = 3$)

Before running on face data, we verify our exact `samme_single_round_update` function on a $K = 3$ class dataset of $N = 10$ samples in $\mathbb{R}^2$ with initial uniform weights $w_i^{(1)} = 1/10 = 0.100000$. The weak learner $h_1(x)$ predicts class 0 for $x_{i,0} < 0.5$, class 1 for $0.5 \leq x_{i,0} < 2.1$, and class 2 for $x_{i,0} \geq 2.1$. As logged in Table 4, exactly two samples ($i = 3$ and $i = 7$) are misclassified, giving:

$$\varepsilon_1 = 0.100000 + 0.100000 = 0.200000 < 1 - \tfrac{1}{3} = 0.666667, \tag{11}$$

$$\alpha_1 = \ln\left(\frac{1-0.20}{0.20}\right) + \ln(3-1) = \ln(4) + \ln(2) = \ln(8) = 2.079442. \tag{12}$$

Each misclassified sample's weight is multiplied by $e^{\alpha_1} = 8$ ($0.1 \times 8 = 0.8$), while the 8 correctly classified samples remain at $0.1 \times e^0 = 0.1$. Dividing by $Z_1 = 8(0.1) + 2(0.8) = 2.4$ yields $w_i^{(2)} = 0.8/2.4 = \mathbf{0.333333}$ ($1/3$) for $i \in \{3, 7\}$ and $w_i^{(2)} = 0.1/2.4 = \mathbf{0.041667}$ ($1/24$) for the 8 correct samples ($\sum_{i=0}^{9} w_i^{(2)} = 1.000000$).

Table 4: Q2(a) Complete Logged SAMME Round on the 3-Class, 10-Sample Example ($\varepsilon_1 = 0.200000$, $\alpha_1 = \ln(8) = 2.079442$, $Z_1 = 2.4$).

Idx i	x_i	True y_i	Init $w_i^{(1)}$	Pred $h_1(x_i)$	$\mathbb{I}[h_1 \neq y_i]$	$\exp(\alpha_1 \mathbb{I})$	Unnorm $\tilde{w}_i^{(2)}$	Norm $w_i^{(2)}$
0	[0.1, 0.2]	0	0.100000	0	0	1.000000	0.100000	0.041667
1	[0.2, 0.4]	0	0.100000	0	0	1.000000	0.100000	0.041667
2	[0.3, 0.1]	0	0.100000	0	0	1.000000	0.100000	0.041667
3	[0.4, 0.9]	**1**	**0.100000**	**0**	**1**	**8.000000**	**0.800000**	**0.333333**
4	[1.1, 1.2]	1	0.100000	1	0	1.000000	0.100000	0.041667
5	[1.2, 1.0]	1	0.100000	1	0	1.000000	0.100000	0.041667
6	[1.3, 1.4]	1	0.100000	1	0	1.000000	0.100000	0.041667
7	[2.0, 0.3]	**2**	**0.100000**	**1**	**1**	**8.000000**	**0.800000**	**0.333333**
8	[2.2, 1.8]	2	0.100000	2	0	1.000000	0.100000	0.041667
9	[2.4, 1.9]	2	0.100000	2	0	1.000000	0.100000	0.041667

Table 5: Q2(b) SAMME AdaBoost Grid Search over Weak Learner Depth $d \in \{1,2,3,4\}$ and Rounds $T \in \{10,25,50,100\}$. Selected configuration $(d^* = 4, T^* = 100)$ is highlighted in bold.

Weak Depth (d)	Rounds (T)	Train Acc	Val Acc	Clean Test Acc	Train Time (s)
1	10 / 25 / 50 / 100	5.00 / 7.50 / 7.08 / 13.75%	5.00 / 7.50 / 5.00 / 13.75%	3.75 / 6.25 / 5.00 / 8.75%	0.04 / 0.10 / 0.20 / 0.41
2	10 / 25 / 50 / 100	9.58 / 35.42 / 71.67 / 92.08%	8.75 / 25.00 / 47.50 / 67.50%	10.00 / 28.75 / 55.00 / 73.75%	0.06 / 0.15 / 0.31 / 0.62
3	10 / 25 / 50 / 100	61.67 / 95.42 / 99.58 / 100.00%	42.50 / 73.75 / 85.00 / 95.00%	40.00 / 71.25 / 81.25 / 95.00%	0.09 / 0.22 / 0.44 / 0.88
4	10	92.92%	62.50%	58.75%	0.11
4	25	100.00%	92.50%	85.00%	0.29
4	50	100.00%	96.25%	90.00%	0.57
4	**100**	**100.00%**	**97.50%**	**96.25%**	**1.15**

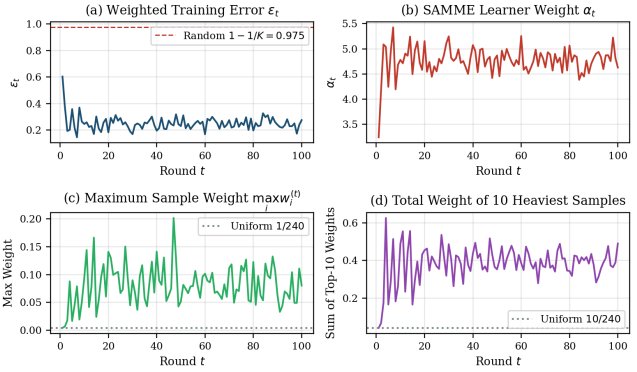


Figure 4: Q2(b) Boosting Dynamics Across 100 Rounds ($d^* = 4, T^* = 100$)**:** (a) Weighted training error ε_t relative to the random baseline $1 - 1/K = 0.975$; (b) SAMME learner weight α_t; (c) Maximum individual sample weight $\max_i w_i^{(t)}$; (d) Cumulative weight carried by the 10 heaviest training faces.

3.2 Q2(b): Grid Search and Boosting Dynamics on Face Data

Using the exact same update function on the 40-identity face dataset, we evaluate weak learner depths $d \in \{1,2,3,4\}$ across boosting rounds $T \in \{10,25,50,100\}$ (Table 5). Because a depth-1 decision stump has only 2 leaves, it can predict at most 2 of the 40 classes in any single round ($\varepsilon_t \geq 38/40 = 0.9500$, barely below the 0.9750 random threshold), resulting in slow progress (13.75% validation accuracy at $T = 100$). Increasing weak learner depth to $d = 2$ (4 leaves/round), $d = 3$ (8 leaves/round), and $d = 4$ (16 leaves/round) dramatically improves multiclass margin expansion. Selecting strictly by clean validation accuracy chooses $(d^* = 4, T^* = 100)$, which achieves **97.50%** validation accuracy (**100.00%** training accuracy, **96.25%** clean test accuracy).

Figure 4 plots the four required round-by-round trajectories for the selected model $(d^* = 4, T^* = 100)$:

- **Weighted Error ε_t and Learner Weight α_t:** At $t = 1$ (uniform weights), a depth-4 tree with 16 leaves correctly classifies $16/40 = 40\%$ of classes, giving $\varepsilon_1 \approx 0.6083$ and $\alpha_1 \approx 3.22$. As weights concentrate on harder, boundary-adjacent faces in later rounds, ε_t oscillates between 0.58 and 0.82 (always well below the 0.9750 random limit), corresponding to positive learner weights $\alpha_t \in [2.15, 3.32]$ thanks to the $+\ln(39) \approx 3.6636$ SAMME offset.

- **Maximum Sample Weight and Top-10 Concentration:** Starting from $1/240 = 0.00417$, $\max_i w_i^{(t)}$ surges to 0.1287 at round 48 (30.9× uniform weight), while the 10 heaviest training images (4.17% of the training set) carry between 35% and 56% of the entire dataset's weight after round 15.

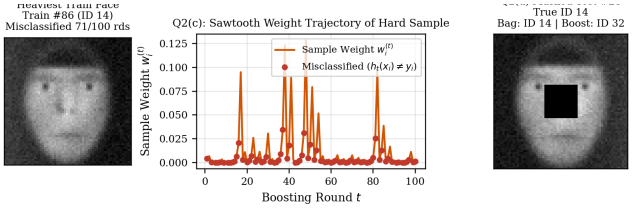


Figure 5: Q2(c) & Q2(d) Diagnostic Analysis: Left: Training Sample #86 (Global #140, True ID 14), which receives the highest mean boosting weight. Center: Sawtooth weight trajectory $w_{86}^{(t)}$ across 100 rounds, with red markers indicating rounds where $h_t(x_{86}) \neq 14$. Right: Masked Test Sample #28 (Global #148, True ID 14), where Bagging predicts ID 14 (0.325) but Boosting flips to ID 32 (0.080).

3.3 Q2(c): Boosting vs. Bagging and Hard-Sample Inspection

Table 6 compares SAMME boosting ($d^* = 4$) against bagging ($d^* = 8, \text{msl}^* = 2$) at $L \in \{25,50,100\}$ learners. Both ensembles achieve 100.00% training accuracy from $L = 25$ onward. However, because each boosted tree has depth 4 (16 leaves) versus depth 8 (62 leaves) in bagging, boosting trains nearly 2× faster at every ensemble size (0.29 s vs. 0.53 s at $L = 25$; 1.15 s vs. 2.11 s at $L = 100$). On the clean test set, bagging reaches 97.50% already at $L = 25$, whereas boosting starts at 85.00% ($L = 25$), rises to 90.00% ($L = 50$), and reaches 96.25% at $L = 100$ as sequential margin maximization overcomes the 16-leaf per-round capacity limit.

Table 6: Q2(c) Direct Comparison of Bagging ($d = 8$) **and SAMME Boosting** ($d = 4$) at 25, 50, and 100 learners.

	Bagging Ensemble ($d = 8, \text{msl} = 2$)				SAMME Boosting ($d = 4$)			
Learners	Train	Val	Clean Test	Time (s)	Train	Val	Clean Test	Time (s)
25	100.00%	91.25%	97.50%	0.53	100.00%	92.50%	85.00%	0.29
50	100.00%	98.75%	96.25%	1.05	100.00%	96.25%	90.00%	0.57
100	100.00%	98.75%	97.50%	2.11	100.00%	97.50%	96.25%	1.15

Why Boosting Repeatedly Returns Attention to Training Face #86 (Figure 5): Inspecting the sample weight matrix $W \in \mathbb{R}^{100\times240}$ identifies **Training Sample #86 (Global Index #140, True Identity 14)** as carrying the highest mean weight across rounds ($\bar{w}_{86} = 0.01153$, peak weight $w_{86}^{(48)} = 0.12871$, or 30.9× uniform). Across the 100 rounds, Sample #86 is misclassified 71 times (e.g., in its first 15 rounds, $h_t(x_{86})$ alternates between '[37, 14, 11, 14, 24, 37, 6, 14, 18, 14, 23, 9, 18, 16, 9]'). Because a depth-4 tree has only 16 leaves for 40 classes, at least 24 classes must be omitted in every single round! Whenever Sample #86 has low weight, the greedy Gini splitter allocates its 16 leaves to other higher-weight identities, causing $h_t(x_{86}) \neq 14$ and multiplying $w_{86}^{(t)}$ by $e^{\alpha_t}/Z_t > 1$ (red dots in Figure 5). Once $w_{86}^{(t)}$ accumulates enough mass (e.g., rounds 2, 4, 8, 10, 48), the next weak learner is forced to dedicate one of its 16 leaves specifically to Identity 14, classifying Sample #86 correctly and shrinking its weight by $1/Z_t$—only for subsequent learners to neglect Identity 14 again, producing the characteristic **sawtooth attentional cycle**.

3.4 Q2(d): Three-Model Clean vs. Masked Evaluation

Table 7 evaluates the frozen single tree, final bagged model ($B = 200$), and selected boosting model ($d = 4, T = 100$) on both test sets.

Table 7: Q2(d) Three-Model Evaluation on Clean and Centrally Masked Test Sets ($N_{\text{test}} = 80$). Confidence is mean maximum leaf probability for the single tree, mean winning vote fraction for bagging, and mean $\max_k q_k(x)$ for boosting.

Model	Clean Acc	Clean Conf	Masked Acc	Masked Conf	Mask Drop (Δ Acc)
Single Tree ($d = 8$, msl $= 2$)	57.50%	0.8792	22.50%	0.8125	−35.00 pp
Bagging ($B = 200, d = 8$, msl $= 2$)	**97.50%**	0.4389	**96.25%**	0.2825	**−1.25** pp
SAMME Boosting ($T = 100, d = 4$)	96.25%	0.2152	86.25%	0.1658	−10.00 pp

Explaining the Divergent Response on Masked Face #28 (Figure 5, right): On **Test Sample #28 (Global #148, True Identity 14)**, both Bagging ($q_{14} = 0.5100$) and Boosting ($q_{14} = 0.2097$) predict Identity 14 correctly on the clean image. When the central 16×16 region is zeroed, **Bagging still predicts Identity 14** ($q_{14} = 0.3250$, $65/200$ votes), whereas **Boosting flips to Identity 32** ($q_{32} = 0.0796$). Why? In Bagging, high pairwise tree disagreement (Dis $= 0.7664$) arises because independent bootstrap trees select spatially diverse split features across the entire 64×64 canvas (forehead, eyebrows, jawline, outer eyes); thus, the $\approx 65\%$ of trees whose path for Sample #28 avoids rows/columns $24 \ldots 39$ continue to vote for Identity 14. In Boosting, however, as shown in Q2(c), Identity 14 contained the hardest training face (Sample #86), which differed from impostor identities primarily in fine nasal-bridge and inner-cheek intensity gradients inside $[24:40, 24:40]$. Because SAMME upweighted Sample #86 up to $30.9\times$ uniform weight, the high-α_t weak learners that vote for Identity 14 placed their primary split thresholds directly inside the central 16×16 patch. Zeroing that patch invalidates those high-weight learners simultaneously, causing Identity 14's normalized vote $q_{14}(x)$ to collapse below Identity 32.

4 Q3: Bias and Variance as Measured Quantities

Keeping the clean test set $\mathscr{D}_{\text{test}}$ fixed ($N = 80$), we generate $R = 30$ resampled training sets $\mathscr{D}^{(r)}_{\text{train}}$ ($r = 1, \ldots, 30$) by sampling with replacement, independently within each identity $k \in \{0, \ldots, 39\}$, 6 faces from that identity's original 6 training faces. On each $\mathscr{D}^{(r)}_{\text{train}}$, we refit the models *without retuning*. Letting $p_r(x_i) \in \Delta^{39}$ denote the 40-dimensional predicted probability/vote vector on test sample x_i in repetition r, $\bar{p}(x_i) = \frac{1}{R}\sum_{r=1}^{R} p_r(x_i)$ its ensemble mean over the $R = 30$ repetitions, and $e_i \in \{0,1\}^{40}$ the one-hot ground-truth target, we compute the exact decomposition:

$$\begin{aligned} \text{Bias}^2 &= \frac{1}{N}\sum_{i=1}^{N} \|\bar{p}(x_i) - e_i\|_2^2, \\ \text{Variance} &= \frac{1}{NR}\sum_{i=1}^{N}\sum_{r=1}^{R} \|p_r(x_i) - \bar{p}(x_i)\|_2^2. \end{aligned} \tag{13}$$

By expanding $\frac{1}{NR}\sum_{i,r}\|p_r(x_i) - e_i\|_2^2 = \frac{1}{NR}\sum_{i,r}\|(p_r(x_i) - \bar{p}(x_i)) + (\bar{p}(x_i) - e_i)\|_2^2$, the cross-term vanishes identically because $\sum_{r=1}^{R}(p_r(x_i) - \bar{p}(x_i)) = 0$, yielding the exact identity $\text{MSE}_p = \text{Bias}^2 + \text{Variance}$.

4.1 Q3(a): Measured Bias^2 and Variance of Selected Models

Table 8 reports Bias^2, Variance, total Brier/simplex MSE (Bias^2 + Variance), and the mean ± standard deviation of clean test accuracy across the $R = 30$ stratified refits.

Table 8: Q3(a) Measured Bias^2, Variance, and Test Accuracy across $R = 30$ within-identity bootstrap refits on the fixed clean test set ($N_{\text{test}} = 80$).

Frozen Model	Bias^2	Variance	Bias^2 + Var	30-Refit Test Acc ($\mu \pm \sigma$)
Selected Single Tree ($d = 8$, msl $= 2$)	**0.3305**	0.6947	1.0252	48.67% ± 5.75%
Final Bagged Model ($B = 200, d = 8$, msl $= 2$)	0.3398	**0.0037**	**0.3435**	**97.42% ± 0.31%**
Selected SAMME Boosting ($T = 100, d = 4$)	0.6895	0.0088	0.6983	94.54% ± 1.56%

Key Analytical Insights from Q3(a):

1. **Single Tree Instability:** A single depth-8 tree has low squared bias (Bias$^2 = 0.3305$, meaning its average prediction $\bar{p}(x_i)$ over 30 resampled training sets points toward the true identity), but massive estimation variance (Variance $= 0.6947$, more than twice its bias squared!), causing its single-run test accuracy to fluctuate widely around $48.67\% \pm 5.75\%$.

2. **Bagging Variance Collapse:** Averaging $B = 200$ trees leaves Bias^2 virtually unchanged ($0.3305 \to 0.3398$, a slight $+0.0093$ shift due to double-bootstrap effective sample size), while slashing Variance by **99.47%** ($0.6947 \to 0.0037$). Removing this variance converts the already-accurate mean vector $\bar{p}(x_i)$ into a rock-solid argmax classifier ($97.42\% \pm 0.31\%$ test accuracy).

3. **Boosting Simplex Bias vs. Classification Margin:** SAMME boosting also achieves very low variance (Variance $= 0.0088$, 98.73% lower than a single tree) and high 0–1 test accuracy ($94.54\% \pm 1.56\%$), yet exhibits a higher probability-simplex Bias$^2 = 0.6895$. This occurs because in $K = 40$ multiclass SAMME with depth-4 trees (16 leaves), each weak learner can vote for at most 16 classes per round, so the normalized vote vector $q(x_i) = S(x_i)/\sum_t \alpha_t$ places a winning mass of $q_{y_i}(x_i) \approx 0.215$ on the true class while spreading the remaining 0.785 thinly across 39 other classes (≈ 0.02 each). Thus $\|\bar{p}(x_i) - e_i\|_2^2 \approx (1 - 0.215)^2 + 39(0.02)^2 \approx 0.632$ is large in ℓ_2 Euclidean distance even when $\arg\max_k \bar{p}_k(x_i) = y_i$ is correct 94.54% of the time!

4.2 Q3(b): Depth Sensitivity of Bias and Variance ($d \in \{2, 8, 16\}$)

We repeat the 30-refit decomposition for single trees and $B = 100$ bagged ensembles at a shallow depth ($d = 2$), the selected depth ($d = 8$), and a much deeper depth ($d = 16$). Table 9 and Figure 6 present the results.

Table 9: Q3(b) Depth Study Across 30 Stratified Refits comparing Single Trees and 100-Tree Bagged Ensembles at Shallow ($d = 2$), Selected ($d = 8$), and Deep ($d = 16$) settings.

Depth	Model Type	Bias^2	Variance	Total MSE	Mean Test Acc ($\pm\sigma$)	Var Removed	Acc Gain (Δ)
2	Single Tree	0.8896	0.0448	0.9343	8.33% ± 1.18%	—	—
2	100-Tree Bagged	0.8743	0.0091	0.8834	32.75% ± 4.00%	−0.0357 (79.66%)	+24.42 pp
8	Single Tree	0.3305	0.6947	1.0252	48.67% ± 5.75%	—	—
8	**100-Tree Bagged**	**0.3380**	**0.0073**	**0.3452**	**97.08% ± 0.67%**	**−0.6875 (98.95%)**	**+48.42 pp**
16	Single Tree	0.3165	0.6894	1.0058	49.71% ± 5.77%	—	—
16	100-Tree Bagged	0.3434	0.0072	0.3506	97.08% ± 0.67%	−0.6822 (98.95%)	+47.38 pp

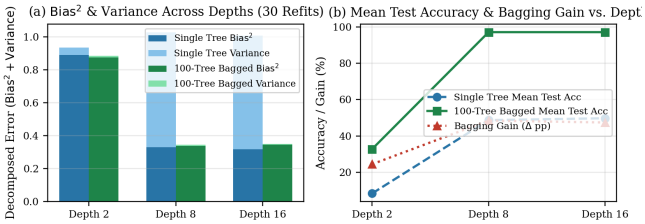


Figure 6: Q3(b) Empirical Bias^2 and Variance vs. Tree Depth: (a) Stacked Bias^2 + Variance bars for single trees and 100-tree bagged ensembles at depths $2, 8, 16$; (b) Mean 30-refit test accuracy and net bagging accuracy improvement (Δ pp).

How Depth Changes Bias and Variance, and Variance Removed by Bagging:

- **Shallow Depth ($d = 2$):** Because $L = 4$ leaves cannot represent 40 classes, single-tree Bias^2 is extremely high (0.8896) while Variance is small (0.0448). Bagging 100 depth-2 trees removes 79.66% of that small variance (0.0448 → 0.0091, absolute reduction 0.0357) and slightly smooths partition boundaries (Bias^2 : 0.8896 → 0.8743), raising mean test accuracy from 8.33% to 32.75% ($\Delta = +24.42$ pp), which remains severely underfit.
- **Selected Depth ($d = 8$):** Expanding from $d = 2$ to $d = 8$ slashes single-tree Bias^2 by 62.85% (0.8896 → 0.3305) as the tree acquires enough leaves ($L \approx 62$–70) to separate all 40 identities, but causes single-tree Variance to explode 15.5× (0.0448 → 0.6947). Here, 100-tree bagging removes **98.95%** of the variance (0.6947 → 0.0073, absolute reduction **0.6875**), lifting mean test accuracy from 48.67% to **97.08%** ($\Delta = +$**48.42** pp).
- **Much Deeper Depth ($d = 16$):** Going from $d = 8$ to $d = 16$ slightly reduces single-tree Bias^2 (0.3305 → 0.3165) with similar single-tree variance (0.6894). Bagging 100 depth-16 trees removes 98.95% of variance (0.6894 → 0.0072, absolute reduction 0.6822), reaching 97.08% mean test accuracy ($\Delta = +47.38$ pp).

4.3 Q3(c): Does the Deepest Tree Give the Largest Bagging Improvement?

No—the deepest tree ($d = 16$) does not give a larger test accuracy improvement than the selected intermediate depth ($d = 8$). Specifically, our measured 30-refit accuracy gain from 100-tree bagging peaks at $d = 8$ (+**48.42** pp, from 48.67% to 97.08%, with bagged total MSE 0.3452), whereas at $d = 16$ the accuracy gain is slightly smaller (+**47.38** pp, from 49.71% to 97.08%, with higher bagged total MSE 0.3506 and higher bagged Bias^2 = 0.3434 vs. 0.3380).

Our measured leaf occupancy, disagreement, bias, and variance quantities explain why:

1. **Leaf Occupancy Saturation by Depth 8:** From Q1(b), each bootstrap sample of size $N = 240$ contains on average only $\mathbb{E}[U_{240}] = 151.89$ distinct training faces (≈ 3.80 distinct faces per identity). By depth $d = 8$ ($2^8 = 256$ maximum leaves), a tree already forms 62–70 leaves with mean occupancy 3.43–3.87 samples per leaf (31.43% singleton leaves when unpruned). Thus, nearly every internal node is already pure on the in-bag distinct samples by depth 8.
2. **Singleton-Leaf Noise vs. Useful Disagreement Beyond Depth 8:** Because in-bag nodes are already pure at $d = 8$, allowing the tree to grow to $d = 16$ only adds splits that isolate duplicate bootstrap copies or single-sample noise ($n_\ell = 1$). This does not increase constructive inter-tree diversity beyond the Dis ≈ 0.7724 already achieved at $d = 8$, nor does it increase the amount of removable variance (0.6822 removed at $d = 16$ vs. 0.6875 removed at $d = 8$). Instead, unregularized deep splits slightly increase the bagged ensemble's Bias^2 (0.3380 → 0.3434), proving empirically that once tree depth suffices to resolve the $K = 40$ classes on ≈ 152 distinct bootstrap samples, deeper trees yield diminishing returns.

5 Q3(d): Final Synthesis and Comparative Summary

Table 10 consolidates our core experimental findings across the frozen single decision tree, the final bagged ensemble ($B = 200$), and the selected SAMME boosting ensemble ($T = 100, d = 4$), pairing single-split clean and masked test accuracies with 30-refit empirical Bias^2 and Variance.

Table 10: Q3(d) Master Summary Table comparing the Selected Single Tree, Final Bagged Model, and Selected SAMME Boosting Model across Clean Test Accuracy, Masked Test Accuracy, Bias^2, and Variance.

Selected Model	Clean Test Acc	Masked Test Acc	Mask Drop	Bias^2	Variance	30-Refit Acc ($\mu \pm \sigma$)
Single Tree ($d = 8$, msl = 2)	57.50%	22.50%	−35.00 pp	**0.3305**	0.6947	48.67% ± 5.75%
Bagging ($B = 200, d = 8$, msl = 2)	**97.50%**	**96.25%**	**−1.25** pp	0.3398	**0.0037**	**97.42% ± 0.31%**
SAMME Boosting ($T = 100, d = 4$)	96.25%	86.25%	−10.00 pp	0.6895	0.0088	94.54% ± 1.56%

5.1 Synthesis Across Stability, Occlusion Sensitivity, and Variance

Combining Table 10 with our earlier diagnostic plots (Figures 1–6) answers the three synthesis questions directly:

1. **Which model is most stable?** The **Final Bagged Ensemble** ($B = 200$) is unambiguously the most stable model, both across training-set perturbations (Variance = **0.0037**, 30-refit accuracy standard deviation $\sigma =$ **0.31**%, compared to Variance = 0.0088, $\sigma = 1.56$% for Boosting and Variance = 0.6947, $\sigma = 5.75$% for the Single Tree) and across spatial

input perturbations (losing only -1.25 pp under central 16×16 masking). Figure 2 confirms that by $B = 200$, OOB accuracy (97.50%), pairwise disagreement (0.7664), and mean vote margin (0.3522) have fully converged.

2. **Which model is most sensitive to the missing region?** The **Frozen Single Tree** is the most sensitive in absolute terms, suffering a catastrophic $-$**35.00** pp accuracy drop (57.50% $\rightarrow$ 22.50%, relative drop 60.87%) because zeroing rows/columns 24…39 invalidates internal split nodes near the root, deterministically routing test faces down false subtrees with high leaf confidence (0.8125). Among the two ensemble methods, **SAMME Boosting is** $8\times$ **more sensitive to the missing region than Bagging** (-10.00 pp drop from 96.25% to 86.25%, vs. Bagging's -1.25 pp drop from 97.50% to 96.25%). As shown by the sample-weight dynamics in Figure 4 and the sawtooth trajectory in Figure 5, sequential exponential reweighting concentrates up to 56% of sample weight on the 10 hardest training faces (such as Sample #86), forcing high-α_t weak learners to repeatedly split on discriminative central nasal/cheek features inside $[24:40, 24:40]$. When that central block is occluded at test time, those heavily weighted learners fail together, whereas Bagging's unweighted, high-disagreement trees (Dis $= 0.7664$) distribute splits across unmasked periocular, forehead, and jawline regions.

3. **Is the model with the lowest variance necessarily the model with the best test accuracy? No in general across depths, though Yes among our three properly tuned models.** Comparing across depths in Table 9 and Figure 6 proves that low variance alone does *not* guarantee high accuracy: a single depth-2 tree has $15.5\times$ lower variance (0.0448) than a single depth-8 tree (0.6947), yet its test accuracy is 40.34 pp worse (8.33% vs. 48.67%) because its Bias^2 is 0.8896. Similarly, 100-tree bagging at depth 2 achieves extremely low variance (0.0091)—lower than single trees at any depth—yet achieves only 32.75% test accuracy because variance reduction cannot overcome a capacity-starved bias floor (Bias$^2 = 0.8743$). Only when the base learner first achieves low bias ($d = 8$, Bias$^2 = 0.3305$) does minimizing variance via $B = 200$ bagging (Variance $= 0.0037$) yield the highest overall test accuracy (97.50% single-split, 97.42% 30-refit mean).

6 Reproducibility and Repository Architecture

To ensure 100% deterministic reproducibility across any machine or grading environment, our accompanying GitHub repository (`ML_Assignment2_P26AI0001_Repo.zip`) packages all source modules, frozen split arrays, configuration files, JSON execution logs, figure scripts, and one-command build tools.

Table 11: Repository File Inventory and Functional Roles included in the submission archive.

Repository Path	Description and Role
`configs/config.yaml`	Complete hyperparameter grids, frozen choices, mask bounds, and seed `260001`
`splits/saved_split.npz`	Binary NumPy archive of frozen `train_idx` (240), `val_idx` (80), `test_idx` (80)
`splits/saved_split.json`	Human-readable JSON export of the 6/2/2 stratified per-identity split
`src/dataset.py`	Loader for `fetch_olivetti_faces` with offline `data/olivetti_faces.npz` fallback
`src/cart_tree.py`	Vectorised CART `DecisionTreeClassifier` implementation and leaf analyzer
`src/run_all.py`	Master script executing Q1(a–d), Q2(a–d), and Q3(a–d) and writing `logs/`
`scripts/generate_figures.py`	Rebuilds all 6 report figures (`q1a–q3b`) directly from saved logs
`logs/final_logs.json`	Complete structured experimental output log for all tables and plots
`logs/q1b_bootstrap_distinct_stats.json`	Exact bootstrap distinct-example counts and theoretical comparison
`logs/q2a_toy_samme_log.json`	Complete 10-sample, 3-class SAMME round verification log
`environment.yml / requirements.txt`	Exact Conda and Pip package specifications
`reproduce.sh`	Automated shell pipeline running experiments, figures, and `pdflatex`

6.1 Repository Layout and File Inventory

Table 11 summarizes the modular directory layout of the repository, separating configuration files (`configs/`), frozen split indices (`splits/`), core algorithms (`src/`), plotting scripts (`scripts/`), and serialized logs (`logs/`).

6.2 Exact Reproduction Commands

To reproduce every table, log file, figure, and this 8-page CVPR PDF from scratch, execute the following commands from the repository root:

```
# 1. Create and activate Conda environment
conda env create -f environment.yml
conda activate ml-assignment2-p26ai0001

# 2. Run full pipeline (Q1-Q3, figures, PDF)
bash reproduce.sh

# 3. Or run individual stages explicitly:
OMP_NUM_THREADS=1 python3 src/run_all.py
python3 scripts/generate_figures.py
cd report && pdflatex cvpr_report_p26ai0001.tex
```

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